

EXP.NO-3	Write a Servlet program that prints "Hello, World!" when accessed through a browser

AIM:

To write a Java Servlet program that demonstrates the use of Servlets by generating a dynamic web page displaying a welcome message for the TechScope product comparison system.

ALGORITHM:

Input:

The user sends an HTTP request by entering a URL in the browser
`http://localhost:8080/TechScopeApp/home`

Servlet Program Flow:

1.Create Servlet Class

Define a Java class named HomeServlet.
Extend the HttpServlet class.
Use the annotation `@WebServlet("/home")` to map the servlet.

2.Handle HTTP Request:

Override the `doGet()` method to handle HTTP GET requests
Inside `doGet()`:
Set response content type to "text/html".
Obtain `PrintWriter` object to send response.
Generate HTML content dynamically.

3.Generate Output:

Display themed content such as:
"Welcome to TechScope"
"Your Smart Product Comparison Hub"

4.Compile the Servlet:

The servlet is compiled automatically by Eclipse into a .class file.

5 Deploy the Servlet:

The project is deployed on Apache Tomcat server using Eclipse.
Servlet mapping is done using annotation, so web.xml is not required.

6. Execute the Servlet:

Start the Tomcat server.
Run the project using "Run on Server".

7. Access via Browser:

Open browser and enter:
<http://localhost:8080/TechScopeApp/home>

CODE:

```
package com.techscope;

import java.io.*;

import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.WebServlet;

@WebServlet("/home")

public class HomeServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

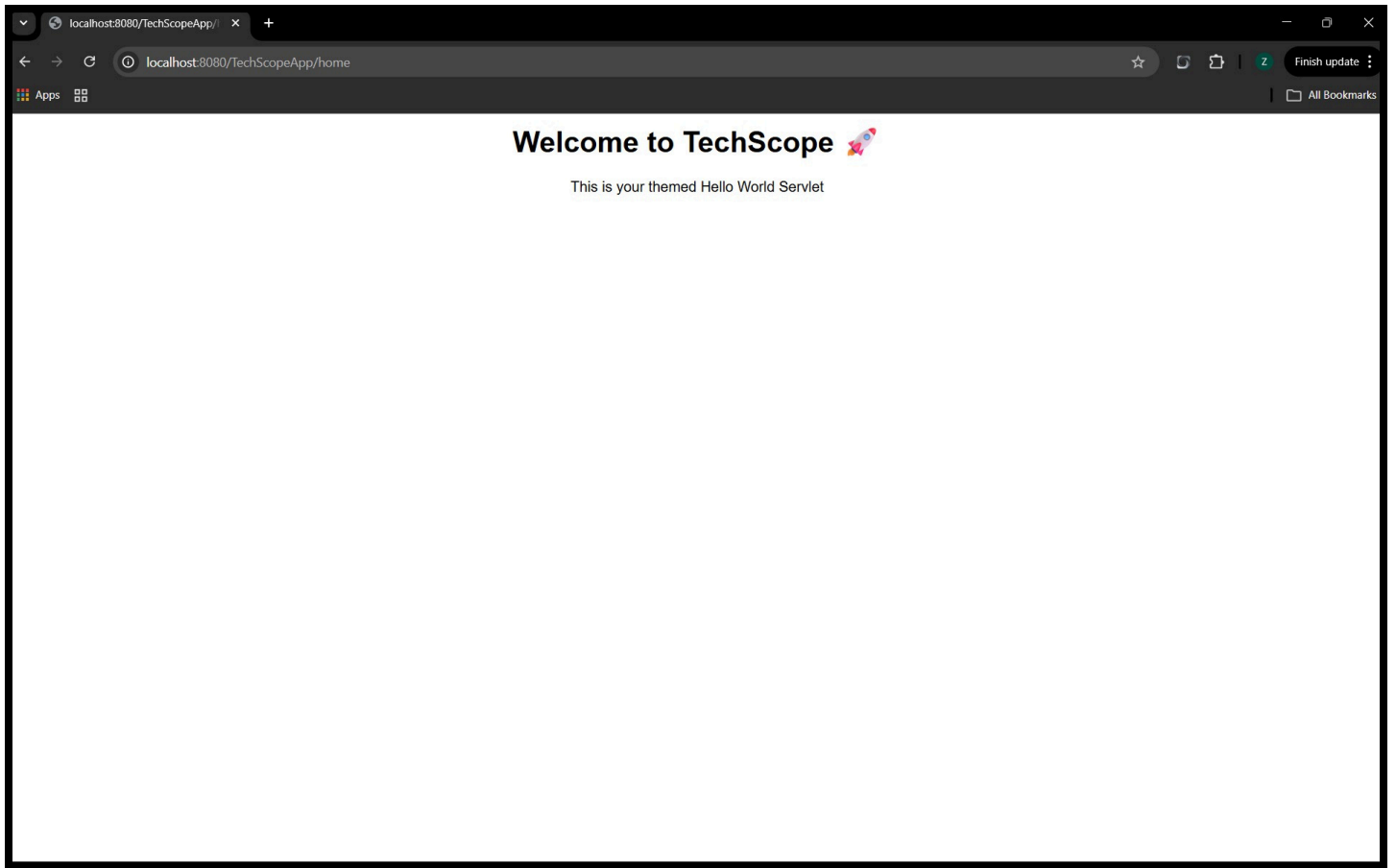
        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html><body style='font-family:Arial; text-align:center;'>");
        out.println("<h1>Welcome to TechScope    </h1>");
        out.println("<p>This is your themed Hello World Servlet</p>");
        out.println("</body></html>");

    }
}
```

OUTPUT:



RESULT:

The Servlet program is executed successfully, and a dynamic web page is displayed in the browser showing the TechScope welcome message, demonstrating the working of Java Servlets.